Greedy Pair-Merge: Background Theory

Bottom-up Agglomerative Graph Coarsening

- Every node starts as its own individual cluster.
- Purely greedy: no undo or backtracking once merged.
- Selects the locally optimal legal pair in every iteration.
- Widely used in CAD, mesh coarsening, and partitioning.

Engine Rules & Specification

Initialization, Contraction, Termination, Cutsize

- Initialization: singletons — every cell is its own cluster.
- Contraction: repeatedly merge the highest-affinity legal pair.
- Termination: stop at K, or when no legal positive-weight pair remains.
- No fake merges: if no valid pair exists, stop even if K is unmet.
- Cutsize: report partition + sum of weights crossing boundaries.

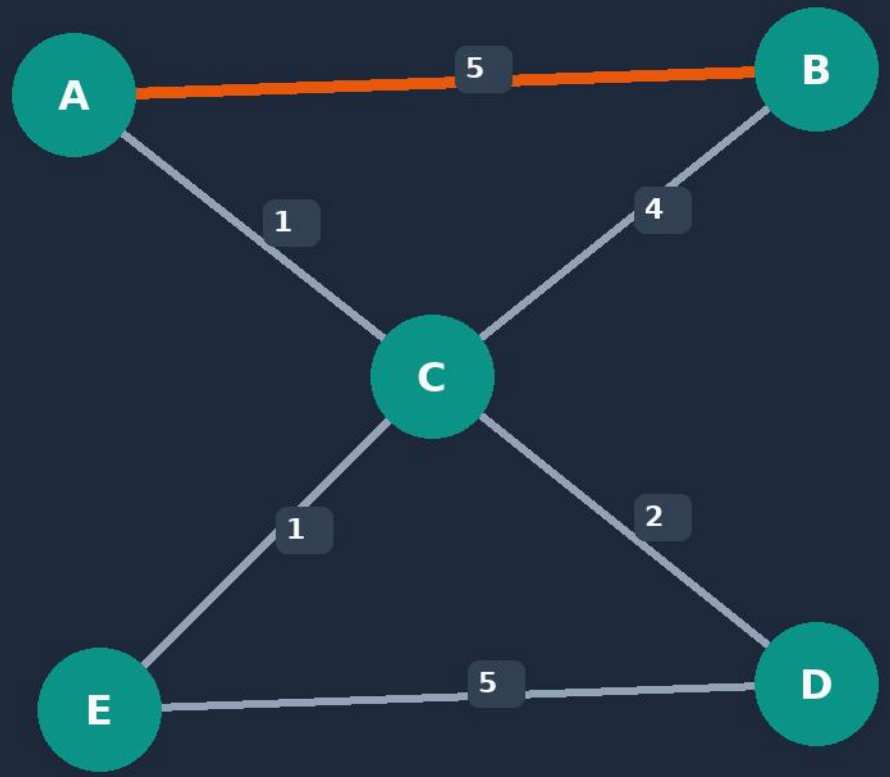
Pair Ranking & Selection Rules

Signal · Rule · Specification

- Affinity: sum current edge weights; parallel connections add.
- Illegal: skip if affinity ≤ 0, or |A| + |B| exceeds capacity.
- Pick: choose the legal pair with maximum affinity.
- Tie-break: lexicographic order of sorted member names joined.
- tiny_graph: A–B vs D–E both 5 → A–B; then AB < D so C is absorbed.

Step 1: Every Node Starts Alone

K=5 Singletons | Target K=2 | Initial Cutsize: 18

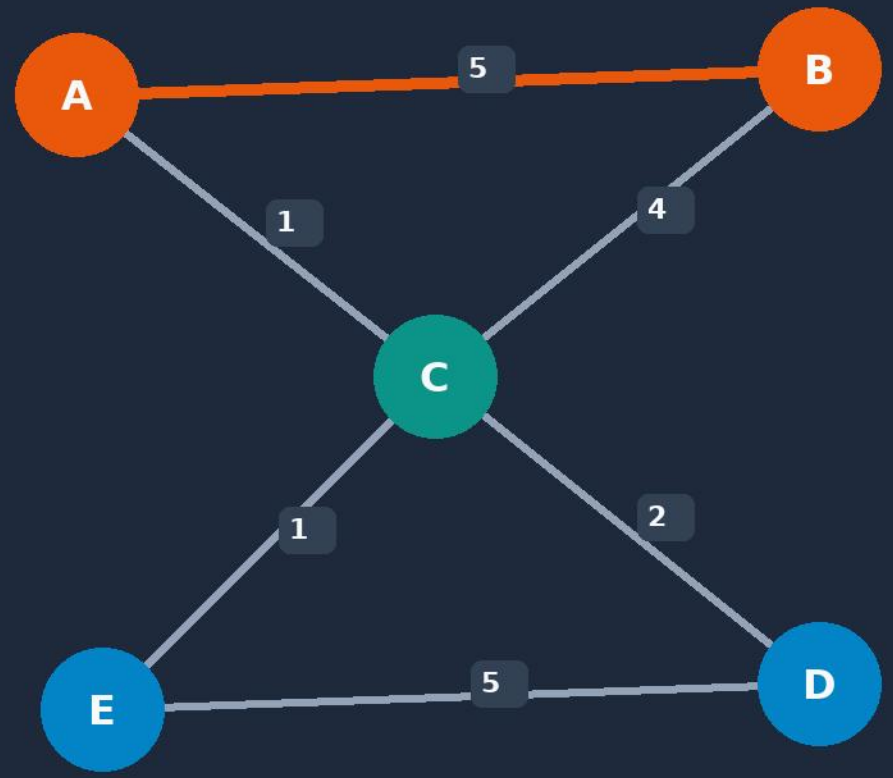


Walkthrough

- Begin with 5 singletons.
- Target K equals 2.
- All-separate cutsize is 18.
- Heaviest edge is A–B at weight 5.

Step 2: Merge 1 (Contract A-B)

Clusters: AB | C | D | E | Cutsize: 13

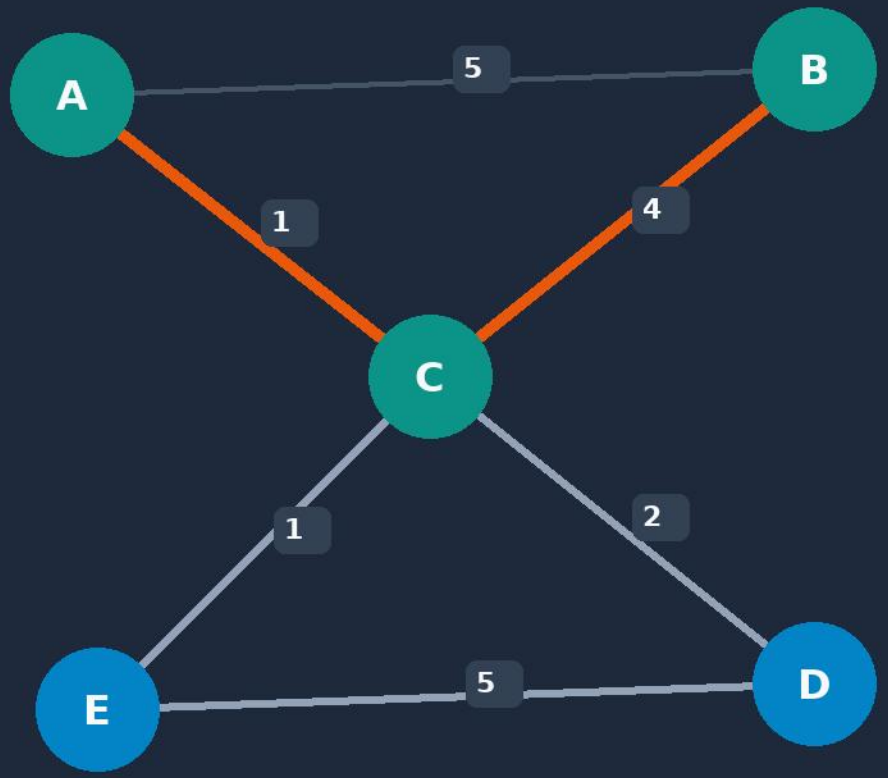


Walkthrough

- Merge A–B at weight 5.
- Protect the strongest connection first.
- Recompute edges into supernode AB.
- Clusters: AB | C | D | E, cutsize 13.

Step 3: Merge 2 (Absorb C into AB)

Clusters: ABC | D | E | Cutsize: 8

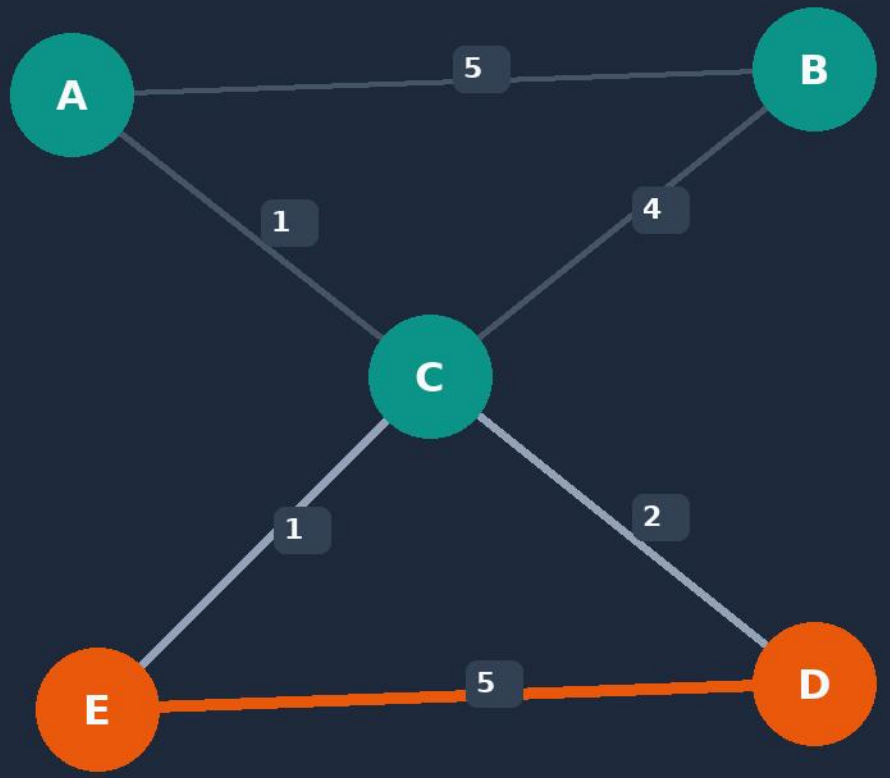


Walkthrough

- C–AB affinity is 1 + 4 = 5.
- That ties D–E at weight 5.
- Lexicographic rule: AB < D, so C merges.
- Clusters: ABC | D | E, cutsize 8.

Step 4: Merge 3 (Contract D-E)

Clusters: ABC | DE | Cutsize: 3 (Target K=2 Reached)

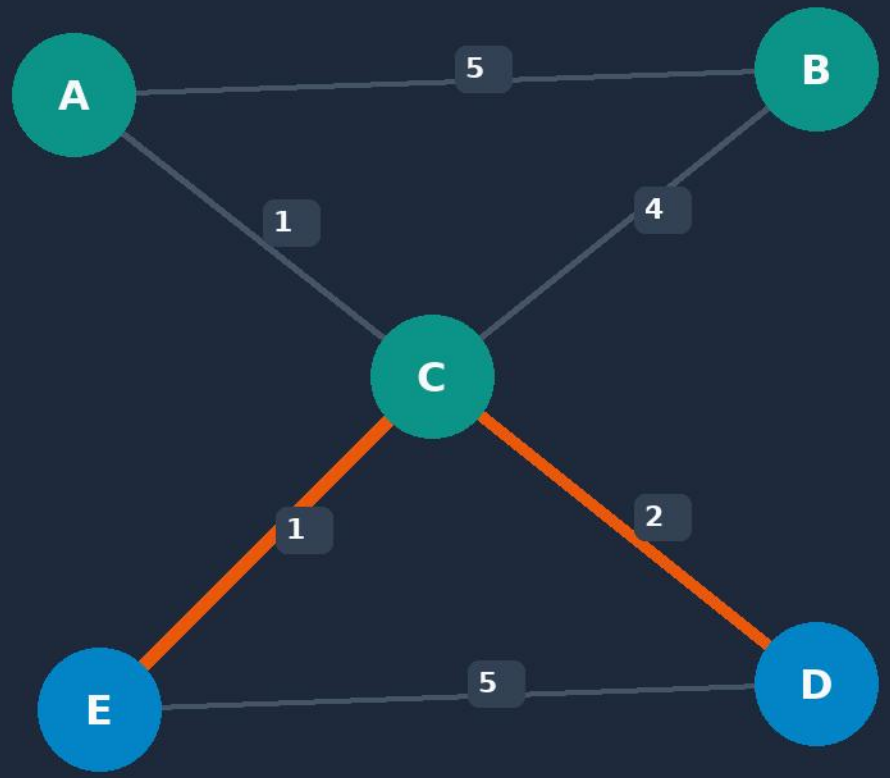


Walkthrough

- Last merge: D–E at weight 5.
- Two clusters: ABC versus DE.
- Target K=2 is reached.
- Only weak bridges remain on the cut.

Step 5: Final Result & Capacity Behavior

Final Partition: ABC | DE | Remaining Cutsize: 3



Walkthrough

- Cut edges: C–D weight 2, C–E weight 1.
- Unconstrained golden: ABC|DE, cut 3.
- Capacity=2 blocks size-3 ABC.
- Cap-2 result: AB|C|DE, cutsize 8.

Critical Implementation Pitfalls

What breaks goldens if the engine is slightly wrong

- Myopic decisions: an attractive first merge can lock in a bad cut.
- Capacity order: check capacity BEFORE ranking, never after.
- Affinity updates: recompute and sum parallel edges after merge.
- Deterministic tie-break: sort combined labels so tests stay stable.
- If capacity blocks every pair, stop — do not invent a merge for K.

Summary: Greedy Pair-Merge

Key Takeaways for CAD & Graph Partitioning

- Fast bottom-up coarsening engine for CAD and mesh problems.
- No backtracking: a local optimum may not equal the global optimum.
- Capacity limits can prevent reaching target cluster count K.
- Correct affinity updates and a stable lex tie-break are required.